[INCH-POUND]
A-A-2932A
October 22, 1997
SUPERSEDING
A-A-2932
December 15, 1995

COMMERCIAL ITEM DESCRIPTION

KEY, HEXAGON SOCKET HEAD SCREW AND KEY SETS SOCKET HEAD

The General Services Administration has authorized the use of this Commercial Item Description for procurement of the Type I key specified in Federal Specification GGG-K-275D for all federal agencies.

1. SCOPE

This Commercial Item Description covers "L" shaped, hexagon socket head, socket head screw keys and sets.

2. CLASSIFICATION

Type I - Short series.
Type II - Long series.

3. SALIENT CHARACTERISTICS

3.1 Material. The key shall be made of high grade alloy steel hardened and tempered to comply with the specified physical and mechanical requirements.

3.2 Dimensions. The key shall conform to the dimension specified for "L" shaped hexagon socket head screw keys in Table 8 of ASME/ANSI B18.3. Measurements across flats and corners shall be taken 1/32 inch from the end on sizes 0.028 inch through 0.109 inch and 1/16 inch from the end on sizes 0.125 inch or larger. For keys with chamfered ends, the measurements shall be made from the end of the chamfer rather than from the end of the key.

3.3 Coating. The key shall be coated with black oxide or phosphate. The coating shall be adherent, smooth, continuous, and free from uncoated areas.

3.4 Sets. The components of sets shall conform to the specified requirements for the individual items.

3.5 Carrying containers. Carrying containers for sets shall be constructed of fabric or plastic. The design shall be a wrap, roll, envelope or pouch. Each carrying container shall be marked in a legible manner with the manufacturer's name and the National Stock Number (NSN) of the item.

3.5.1 Carrying container material.

3.5.1.1 Fabric. The fabric shall be cloth duck material of sufficient grade and thickness to properly protect and retain the keys. The fabric shall be treated to resist water, mildew, and fungus and shall be olive drab.

Comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Center (6FET), Building 6, 1500 E. Bannister Road, Kansas City, MO 64131.

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3.5.1.2 Plastic. The plastic shall be vinyl plastic and shall be a minimum of 18 gage thick, color optional. The pockets shall be a minimum of 12 gage thick with a matte finish (40 to 50% clear).

3.5.2 Carrying container design.

3.5.2.1 Wrap. The design of the wrap shall be appropriate to securely contain the set components. The wrap shall contain each different size key in an individual pocket marked with the key size and shall have no empty pockets. All keys shall be easily removable and replaceable in the pockets. The pocket size shall be appropriate to securely contain the key. When stitching is used, it shall be of the lock stitch type. Wraps shall have one or more snap type fastener(s) for securely closing the wrap. The fastener(s) shall be easily opened and closed by hand.

3.5.2.2 Roll. The design of the roll shall appropriate to securely contain the set components. The roll shall contain each different size key in an individual pocket marked with the key size and shall have no empty pockets. All keys shall be easily removable and replaceable in the pockets. The pocket size shall be appropriate to securely contain the key. When stitching is used, it shall be of the lock stitch type. Rolls shall have a cloth tape or webbed strap attached to the back surface of the roll bag. A buckle type closure system shall be used to provide a means of retaining the tool roll in a rolled position.

3.5.2.3 Envelope. The size and design of the envelope shall be appropriate to securely contain the set components. The envelope shall have one pocket with one or more snap type fasteners for securing the envelope. The fastener(s) shall be easily opened and closed by hand.

3.5.2.4 Pouch. The pouch shall comply with the requirements of paragraph 3.5.2.3, but shall have separate pockets for each different size key. The pocket size shall be appropriate to securely contain the keys.

3.6 Hardness. Keys sized 3/8 inch and larger shall have a minimum hardness of 45 on the Rockwell C scale and a maximum hardness of 60 on the Rockwell C scale. Keys sized smaller than 3/8 inch shall have a minimum hardness of 48 on the Rockwell C scale and a maximum hardness of 60 on the Rockwell C scale. When required to be tested, each sample key size 0.094 inch and larger shall be tested in accordance with ASTM E18. Each sample key size smaller than 0.094 inch shall be tested for hardness with a Knoop indenter in accordance with ASTM E384 or by the Rockwell superficial hardness test in accordance with ASTM E18 .

3.7 Torsion load. The key shall conform to the torsion load specified in Table 1. Suitable sockets shall be used to provide support and necessary strength for the test load applied. The key shall show no evidence of cracks, rounding of corners, or other permanent deformation.

Table 1. Torsional Loads

NOMINAL KEY SIZE (INCHED)	TORSIONAL LOAD (INCH-POUNDS)
0.028	0.94
0.035	1.80
0.050	5.25
1/16	10.25
5/64	20.0
3/32	35.0
7/64	55.0
1/8	82.0
9/64	118.0
5/32	160.0

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3/16	278.0
7/32	440.0
¼	665.0
5/16	1275.0
3/8	2200.0
7/16	3500.0
½	5200.0
9/16	6500.0
5/8	9000.0
¾	15500.0
7/8	24600.0
1	28800.0

3.8 Identification marking. Each key (including all items within a set or kit) having a width across the flats larger than 1/8 inch shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture. Each key having a width across the flats of 1/8 inch or smaller need not be marked. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.9 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Coatings and finishes shall be smooth, abherent, continuous, and not stained or discolored. External surfaces shall be free from tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects(e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety or appearance.

4. REGULATORY REQUIREMENTS

4.1 Recovered Materials. The offeror/contractor is encouraged to use recovered materials to maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall provide certification and maintain substantiating evidence that all items being offered for acceptance meet the requirements of the Commercial Item Description and conform to the manufacturer's drawings. Specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Inspection responsibility. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

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5.3 Bid samples(s). When a bid sample requirement is specified in the solicitation, the bid samples(s) shall be inspected for all salient characteristics. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article samples(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING

Requirements for preservation, packing, packaging, and marking of containers shall be as specified in the contact or order.

7. NOTES

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Address for obtaining copies of referenced documents.

Federal Acquisition Regulation (FAR): Government Printing Office. Superintendent of Documents, Washington, DC 20402-9371.

ASTM: American Society for Testing Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

ASME/ANSI B18.3: American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, NY 10017.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

Federal Specifications: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and shall include the following information in procurements documents.

(a) Title, number, and date of this Commercial Item Description.
(b) The appropriate NSN.
(c) The type and size of key, or the type and size of each key within the set.
(d) Acceptable fabric for carrying containers of sets when applicable.
(e) Acceptable design for carrying containers of sets when applicable.
(f) If bid samples are required and the number of samples required.
(g) If first article samples are required the number of samples required.
(h) If lot inspection and the applicable lot sampling requirements.

For example, the following typical lot sampling requirements would be specified in the contract:

Sampling for Inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

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Inspection. Each sample key shall be examined and tested for all salient characteristics. The inspection Level shall be Level I with an Acceptable Quality Level of 4.0 expressed in percent defective.

7.3 National Stock Numbers (NSN's). The following NSN's are associated with this CID. This list may not be indicative of all possible NSN's associated with this CID.

NSN	TYPE	SIZE
5120-00-555-2639	I	0.028
5120-00-198-5400	I	0.035
5120-00-198-5401	I	0.050
5120-00-198-5398	I	1/16
5120-00-224-2504	I	5/64
5120-00-242-7410	I	3/32
5120-00-889-2162	I	7/64
5120-00-240-5292	I	1/8
5120-00-889-2163	I	9/64
5120-00-198-5392	I	5/32
5120-00-240-5300	I	3/16
5120-00-242-7411	I	7/32
5120-00-224-4659	I	1/4
5120-00-240-5274	I	5/16
5120-00-198-5390	I	3/8
5120-00-240-5277	I	7/16
5120-00-198-5391	I	1/2
5120-00-240-5268	I	9/16
5120-00-224-2510	I	5/8
5120-00-222-1489	I	3/4
5120-00-198-5387	I	1
5120-00-596-1236	II	0.035
5120-00-541-5556	II	0.050
5120-00-293-1476	II	1/16
5120-00-293-0080	II	5/64
5120-00-221-8036	II	3/32
5120-00-951-6589	II	7/64
5120-00-984-0247	II	9/64
5120-00-198-5413	II	5/32
5120-00-198-5410	II	3/16
5120-00-241-3179	II	7/32
5120-00-241-3180	II	1/4
5120-00-198-5409	II	5/16
5120-00-198-5406	II	3/8
5120-00-198-5407	II	1/2
5120-00-198-5404	II	9/16
5120-00-198-5405	II	5/8
5120-00-725-7451	II	7/16

Military Custodians:
Army - GL
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Preparing Activity:
GSA-FSS

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